Demo for Data and Network Security report

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Abstract—As mobile devices become ubiquitous, the resulting cellular telemetry forms an indelible human footprint. Modern forensic frameworks such as SURE and MoCo exploit behavioral metadata to bridge the gap between anonymity and identity, rendering traditional anonymization techniques like k-anonymity ineffective. This report analyzes these tracking frameworks and proposes the adoption of a parallel radio network based on the Meshtastic protocol as the only robust mitigation. By bypassing centralized cellular operator data systems, we demonstrate that operational security can be maintained even against deep-learning-driven surveillance.

Index Terms—Machine Learning, Deep Learning, Meshtastic, User Re-identification, Operational Security, LoRa.

1 Introduction

1.1 The Battlefield of Data

In the modern digital landscape, data has become a literal battlefield where every wireless transmission leaves a trace [2]. As long as a mobile connection is maintained, a device generates a continuous stream of telemetry that forms an indelible human footprint [2]. This footprint is captured passively by Cellular Network Operator Data Systems, which aggregate vast quantities of metadata regarding service requests, data exchanges, and handovers between tracking areas [2], [3]. Through their smartphones and the operator network infrastructure, users interact with nearby base stations for communication services such as phone calls, text messages, and Internet browsing. All of these activities are aggregated and stored in Business Support Systems (BSS) and Operating Support Systems (OSS), which serve as the foundation for operator data management [2].

1.2 The Problem—Behavior as Identity

The fundamental challenge is that human habits do not change when a name is erased [2]. Modern forensic frameworks like SURE (Smartphone Users RE-identification) and MoCo (Mobile Contact Detection) treat behavior as identity [2], [3]. By extracting "pure behavioral signatures" from sequences of Base Station (BS) associations and traffic levels, these frameworks can uniquely identify a user [2], [3]. Research indicates that over 60% of user traffic is concentrated on just two antennas—the "Top 2 BS" rule—which serves as a spatial anchor for identifying a target's home or workplace [2].

1.3 The Illusion of Anonymity

Traditional security methods, such as k-anonymity and the suppression of identifiers like IMSI, are "blunt weapons" against deep learning [2], [3]. These methods fail because they only obscure identifying fields while leaving underlying spatio-temporal metadata unchanged [2], [3]. Dual-encoder architectures can mathematically bridge the gap between an anonymous index and a known identity with AUC scores above 0.9 using only 4 days of data [2].

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1.4 Project Objectives

This report demonstrates that as long as a user remains on a cellular grid, full operational disclosure is a mathematical certainty [2], [3]. We propose bypassing the cellular operator data system entirely by deploying a parallel radio network built on the Meshtastic protocol [6].

2 Background

2.1 Deep Learning Networks

Complex computational problems can be addressed through various methodologies, prominent among which is Deep Learning. This paradigm is fundamentally rooted in mathematical optimization. Consider a predictive model defined as:

$$\begin{aligned} &f_\theta : \mathbb{R}^K \to \mathbb{R}^m \\ &\forall x \in \mathbb{R}^K, \Theta(x) \in \mathbb{R}^m \end{aligned} \tag{1}$$

The objective of training is to optimize the internal parameters—specifically the weights and biases of the neurons—to maximize the model's predictive accuracy.

This optimization process is systematically executed through the following iterative phases:

- Initialization: The network parameters are initially set using randomized values or predefined distribution strategies.
- Forward Propagation: An input vector x is processed through the mapping function to generate a predicted output, denoted as $\hat{y}$ (commonly referred to as the forward pass).
- Error Evaluation: The discrepancy between the predicted output and the ground-truth label y is quantified by evaluating a Loss function $\mathbf{L}$ (defined in Equation 2).
- Backpropagation and Optimization: Utilizing the backpropagation algorithm, the gradient of the loss function is computed with respect to each network weight via the chain rule. The model parameters are subsequently adjusted in the negative gradient direction—a process known as gradient descent—effectively minimizing the objective function.

$$\hat{y} = \Theta(x) \\ L(x) = y - \hat{y} \tag{2}$$

While the mathematical formulation of the loss function varies depending on the specific task architecture, its underlying objective remains constant: the resulting scalar serves as a performance metric utilized by optimization algorithms to converge toward the global minimum[1].

The foundational principles detailed above describe standard Machine Learning. The distinction between classical Machine Learning and Deep Learning primarily lies in structural complexity; standard Machine Learning models typically rely on shallow architectures, whereas Deep Learning utilizes deep neural networks characterized by multiple successive hidden layers.

Formally, $\text{Deep Learning} \subset \text{Machine Learning}$. While both paradigms maintain distinct domains of utility, this report focuses primarily on the deployment and behavior of Deep Learning architectures.

2.2 Base Station

This infrastructure comprises cellular base stations maintained by telecommunication service providers. These stations aggregate vast quantities of telemetry and location data which, despite the application of anonymization techniques, remain vulnerable to re-identification attacks orchestrated via deep learning models. Consequently, any adversary possessing sufficient computational resources could exploit these vulnerabilities to execute large-scale mass surveillance[2].

2.3 The SURE Framework (User Re-identification)

The SURE (Smartphone Users RE-identification) framework [2] is a deep learning model designed to accurately classify whether two input network traces belong to the same user. It utilizes a parallel dual-encoder architecture to process anonymous and known-identity traces independently, preventing data contamination before mathematical comparison. The framework consists of four components: Trace Encoder, Matching Network, Fusion Network, and Classifier. The Trace Encoder performs spatio-temporal embedding, which vectorizes four key privacy-impacting features: Base Station (BS) IDs, association duration, and discretized levels of uplink and downlink traffic [2]. The Matching Network then computes cross-attention weight scores between sequences of different days, capturing behavioral similarities across time. Extensive evaluations on a large-scale dataset of 10,000 users over four months (comprising more than 200 million network access records from 39,032 BSs) show that SURE achieves an AUC of 0.941, an F1-score of 0.869, and a precision of 0.807 [2]. Notably, even with a data window as narrow as four days, SURE maintains AUC scores of 0.89 with fusion features. Further analysis reveals that older and male users exhibit higher susceptibility to re-identification risks due to more consistent behavioral patterns [2].

1. The parameter configuration that minimizes the loss value.
2. Amnesty International: An Easy Guide to Mass Surveillance

2.4 The MoCo Framework (Contact Detection)

MoCo (Mobile Contact Detection) [3] focuses on identifying users moving in "lockstep," such as those co-traveling in the same vehicle. The framework addresses the problem of detecting mobile contact events between urban users who exhibit similar spatio-temporal traveling behaviors, leveraging cellular signaling traces of the form $\langle$user ID, BS ID, timestamp$\rangle$ [3].

MoCo employs a three-stage pipeline:

- Data Denoising: The DBSCAN clustering algorithm is applied to remove noise from remote BS connections that do not reflect actual user locations [3].
- Spatio-Temporal Filter (ST-Filter): This module eliminates unlikely mobile contact trajectory pairs in both spatial and temporal domains, typically requiring $\geq 60\%$ spatial similarity to retain candidate pairs [3].
- Detection Learning Network: An LSTM-based spatial branch captures correlations among candidate trajectories using quadkey-encoded BS sequences, while an MLP-based temporal branch learns time-step-dependent patterns from one-hot encoded trajectory vectors [3].

The model is trained using cross-entropy loss and outputs a binary mobile contact classification. Extensive experiments on real-world data (241 user trajectories spanning 211 hours and 6,204 km) demonstrate that MoCo achieves an average F1-score of 87.13%, significantly outperforming classical methods such as S-LCSS and ST-LCSS, as well as deep learning baselines including DPLink and DeepTUL [3].

2.5 Vulnerabilities in Self-Supervised Learning (SSL)

While powerful, these frameworks are susceptible to backdoor attacks [4], [7]. An adversary can poison as little as 0.5% of unlabeled training data with a rigid "trigger" patch, causing the model to associate that trigger with a target category at test time [7], [8]. Recent work such as DIFT [8] proposes defenses against data poisoning in contrastive learning, while studies on adversarial susceptibility [4] demonstrate that contrastive self-supervised models are inherently more vulnerable to adversarial perturbations than their supervised counterparts.

3 Proposed Approach: Meshtastic Parallel Network

3.1 Technical Architecture

We propose Meshtastic, a decentralized mesh network utilizing the LoRa physical layer and Chirp Spread Spectrum (CSS) [6]. Signals consist of frequency-sweeping "chirps" with a 16-up-chirp Preamble for synchronization and a specific Sync Word (0x2B) to distinguish the network [6].

3.2 Decentralized Metadata Protection

The primary strength is the total bypass of Cellular Operator Data Systems [6]. In a mesh architecture, there is no central database logging "Top 2 BS" anchors or hourly traffic windows—the features required by SURE and MoCo for re-identification [2], [3], [6]. The SURE framework relies on extracting BS ID sequences, association durations, and traffic levels from centralized operator BSS/OSS databases

[2]. Similarly, MoCo requires access to cellular signaling traces of the form ⟨user ID, BS ID, timestamp⟩ stored by operators [3]. By removing the centralized collection point entirely, Meshtastic eliminates the data source that makes these attacks possible.

3.3 Encryption and Routing

Meshtastic payloads are encrypted using AES-128 shared-key encryption [6]. Only nodes with the correct channel key can interpret data, while decentralized "hop limits" prevent the creation of a trackable historical footprint [6].

3.4 Justification of Selection

The choice of a parallel radio network over software-side mitigations is motivated by the fundamental limitations of the latter. Approaches such as noise injection and differential privacy attempt to obscure behavioral patterns at the data level; however, they suffer from an inherent trade-off: sufficient noise to defeat re-identification renders the data useless for legitimate analytics [2]. The SURE paper demonstrates that even with k-anonymity (identifier suppression) applied, AUC scores remain above 0.9, indicating that anonymization alone is insufficient [2]. Moreover, techniques like ASTRA [1] and RoCL [5] harden the model against adversarial perturbations but do not address the root cause: the centralized logging of spatio-temporal metadata by cellular operators. Meshtastic addresses this root cause by operating on an entirely separate radio infrastructure, ensuring that no cellular operator ever collects the behavioral traces in the first place.

4 SWOT Analysis

To rigorously evaluate the position of the proposed Meshtastic parallel network against existing cellular infrastructure, we conduct a SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis. This comparative framework establishes the credibility of our proposal by systematically examining both internal attributes and external factors.

TABLE 1
SWOT Analysis: Meshtastic Parallel Network vs. Cellular

	Internal	External
Strengths	No centralized operator logs; metadata privacy; AES encryption [3], [6].	Opportunities: High-privacy operational demand; resistance to state-level AI [6].
Weaknesses	Lower data rates; limited propagation range of LoRa [6].	Threats: Signal triangulation; future RF fingerprinting evolutions [6].

Strengths. The most significant internal advantage is the complete absence of a centralized operator data system. As demonstrated by SURE [2], re-identification attacks require access to BS association sequences and traffic usage patterns stored in BSS/OSS databases. By removing this centralized collection point, Meshtastic eliminates the "Top 2 BS" spatial anchors and hourly traffic windows that enable behavioral fingerprinting. Additionally, AES-128 shared-key encryption ensures payload confidentiality at the application layer [6].

Weaknesses. LoRa operates in the sub-GHz ISM bands with data rates limited to approximately 5.47 kbps at the highest spreading factor, making it unsuitable for high-bandwidth applications such as video streaming or large file transfers [6]. The propagation range, while superior to WiFi in line-of-sight conditions, is constrained in dense urban environments with significant signal attenuation. Furthermore, the mesh topology introduces latency with each additional hop.

Opportunities. The increasing demand for operational security in privacy-sensitive domains creates a growing market for off-grid communication solutions. Recent data breaches at major operators (AT&T, T-Mobile, US Cellular) [2] have highlighted the vulnerability of centralized data repositories, strengthening the case for decentralized alternatives. The resistance to state-level AI surveillance represents a significant advantage for journalists, activists, and organizations operating in adversarial environments.

Threats. While Meshtastic avoids cellular metadata collection, it remains potentially vulnerable to RF-level attacks. Signal triangulation using directional antennas could determine the physical location of transmitting nodes. Future advances in RF fingerprinting—which identify unique hardware characteristics of individual radio transmitters—could theoretically enable device-level identification without requiring centralized metadata logs. Additionally, intentional signal jamming in the LoRa frequency bands could disrupt mesh network operations.

5 Evaluation

5.1 Proposed Experimental Setup

We propose comparing a control group on a cellular network against a test group using Meshtastic over 7 days [2], [6]. We will apply the SURE architecture to match anonymous traces back to a leaked identity database [2]. The experimental design follows the methodology established in the original SURE evaluation: the control group's cellular traces will be collected via operator BSS/OSS in the standard format (user ID, BS ID, timestamps, uplink/downlink traffic), while the test group will communicate exclusively through the Meshtastic mesh network. Both groups will perform equivalent daily routines to ensure behavioral comparability. The SURE model, pre-trained on known cellular trace data, will then attempt to re-identify users in both groups.

5.2 Performance Metrics

Success will be measured via AUC, F1-Score, and Precision-Recall [2], [3]. A successful mitigation will show the Meshtastic group's re-identification AUC dropping to near-random levels (≈ 0.5) compared to the control group's ≥ 0.9 [2]. Additionally, we will apply MoCo's contact detection pipeline [3] to evaluate whether co-traveling users within the Meshtastic group can still be identified through their movement patterns. The expected outcome is that, without access to cellular signaling traces, MoCo's F1-score should degrade significantly from the reported 87.13% baseline.

6 Related Work

6.1 The Collapse of K-Anonymity

Traditional anonymization relies on the removal or suppression of direct identifiers such as IMSI numbers, phone numbers, and subscriber names. However, research has demonstrated that these techniques are fundamentally insufficient when spatial behavior remains stable [2]. The SURE framework shows that even with k-anonymity applied to shared datasets, the underlying spatio-temporal metadata—BS associations, traffic patterns, and temporal routines—persists unchanged, enabling re-identification with AUC scores exceeding 0.9 [2].

A critical finding from the SURE study is that user demographics significantly impact re-identification susceptibility. Male users and those aged 45 and above exhibit the highest re-identification risks (AUC scores of 0.95 for males and above 0.93 for users over 45), attributable to their more consistent and predictable logistical routines [2]. Conversely, users aged 18–24 show marginally lower risk, likely due to more variable mobility patterns. These findings demonstrate that the "Top 2 BS" spatial anchor, where approximately 60% of a user's traffic concentrates on just two base stations, provides a near-deterministic fingerprint of home and workplace locations that survives any identifier-level anonymization [2].

Furthermore, MoCo [3] extends the threat beyond individual re-identification to social graph reconstruction. By detecting co-traveling patterns with an F1-score of 87.13%, MoCo can map a target's "inner circle"—the set of users who regularly share transportation routes—using only cellular signaling traces [3]. This compound threat renders k-anonymity not only insufficient for identity protection but also incapable of shielding social associations.

6.2 Alternative SSL Defenses

Several recent works have proposed defensive techniques against adversarial attacks on self-supervised learning (SSL) models. ASTRA (Adversarial Self-Supervised Training with Adaptive-Attacks) [1] introduces an adaptive adversarial training procedure that generates perturbations during the pre-training phase, improving robustness against adversarial inputs. Similarly, RoCL (Robust Contrastive Learning) [5] incorporates adversarial examples directly into the contrastive learning objective, training the encoder to produce consistent representations even under input perturbations.

While these defenses are effective against model-level adversarial attacks, they do not address the root cause of the tracking threat analyzed in this report: the centralized logging of spatio-temporal metadata by cellular operators [2], [3]. ASTRA and RoCL harden the learning pipeline but leave the data collection infrastructure untouched. An adversary with access to operator BSS/OSS databases can simply train new models on the same raw metadata, bypassing any model-side defense. Additionally, DIFT [8] proposes filtering mechanisms against data poisoning backdoor attacks in contrastive learning, but this again operates at the model level rather than addressing data collection.

This fundamental limitation motivates our proposed approach: rather than defending the model, we eliminate the centralized data source by deploying a parallel communication network that never generates the metadata exploited by SURE and MoCo.

7 Conclusions

7.1 Summary of Findings

As long as cellular connections are maintained, full operational disclosure is inevitable due to behavioral predictability [2], [3]. The analysis presented in this report leads to the following key conclusions:

- Behavioral metadata is identity. The SURE framework demonstrates that BS association patterns and traffic usage create unique behavioral fingerprints, achieving re-identification AUC scores of 0.941 on datasets of 10,000 users, even with k-anonymity applied [2].
- Social graphs are exposed. MoCo extends the threat from individual identification to social network reconstruction, detecting co-traveling contacts with an F1-score of 87.13% using only cellular signaling traces [3].
- Software mitigations are insufficient. Traditional anonymization techniques and model-hardening approaches (ASTRA, RoCL) fail to address the root cause: centralized metadata collection by cellular operators [1], [5].
- Physical-layer separation is necessary. The Meshtastic parallel radio network eliminates the data source exploited by both SURE and MoCo by operating on a completely independent infrastructure with no centralized logging [6].

Parallel radio networks like Meshtastic are not just alternatives but technical necessities for operating outside the "Battlefield of Data" [2], [6].

7.2 Future Extensions

Several directions for future work emerge from this analysis:

- Multi-hop scalability. Investigating the performance of Meshtastic mesh networks at scale, particularly the impact of increasing hop counts on latency, reliability, and metadata leakage through traffic analysis.
- Hybrid encrypted mesh systems. Combining Meshtastic's LoRa backbone with short-range protocols (e.g., BLE mesh) to increase bandwidth for local communications while maintaining the metadata-free properties of the long-range network.
- RF fingerprinting resistance. Developing countermeasures against emerging RF fingerprinting techniques that could potentially identify individual LoRa transmitters based on hardware-specific signal characteristics.
- Experimental validation. Conducting the proposed evaluation framework with real user groups to empirically quantify the re-identification risk reduction achieved by Meshtastic compared to cellular baselines.

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